MH1803 Midterm 2 CHEATSHEET MODE: Strict Scope Only

Boundary: Ch 2 §2.4 → Ch 4 §4.4 only. **Out of scope:** conservative fields, potentials, exact differentials, flux, Stokes, divergence theorem.

Double-sided landscape A4.

div / curl / grad / ∇^2?	Helix r(t)?	Surface area?	$\int$F·dr or work?	$\iiint$ or volume?	$\oint$ circulation?
→ Operators (Col 1)	→ FS Chain (Col 1 btm)	→ Monge (Col 2 btm)	→ Line Int (Col 2–3)	→ Triple Int (Col 2)	→ Circ (Col 3)

SCOPE · PRIORITY · EXAM ORDER

In scope: triple integrals, cyl/sph setup, curves, helix/FS, surface area, grad/div/curl/Laplacian, line integrals, circulation.

1. Operators · 2. Helix / FS · 3. Surface area · 4. Line integrals · 5. Triple integrals

Weak-student rule: do not spread effort evenly. Make operator identities, helix template, Monge formula and line-integral workflow nearly automatic first.

CORE OPERATORS · CARTESIAN

$\nabla = \mathbf{i}\,\partial_x + \mathbf{j}\,\partial_y + \mathbf{k}\,\partial_z$

$\nabla f = (f_x, f_y, f_z)$ → vector output

$\nabla\cdot\mathbf{F} = \partial_x F_1 + \partial_y F_2 + \partial_z F_3$ → scalar output

$$\nabla\times\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ F_1 & F_2 & F_3 \end{vmatrix}$$ → middle term sign flips

$\nabla^2 f = f_{xx} + f_{yy} + f_{zz}$ → sum all pure 2nd derivs

Always: $\nabla\times(\nabla f) = \mathbf{0}$, $\nabla\cdot(\nabla\times\mathbf{F}) = 0$.

Product rules: $\nabla(uv) = u\nabla v + v\nabla u$. $\nabla\cdot(u\mathbf{F}) = u\nabla\cdot\mathbf{F} + \mathbf{F}\cdot\nabla u$. $\nabla\times(u\mathbf{F}) = \nabla u\times\mathbf{F} + u\nabla\times\mathbf{F}$

Cross-product rule: $\nabla\times(\mathbf{A}\times\mathbf{B}) = \mathbf{A}(\nabla\cdot\mathbf{B}) - \mathbf{B}(\nabla\cdot\mathbf{A}) + (\mathbf{B}\cdot\nabla)\mathbf{A} - (\mathbf{A}\cdot\nabla)\mathbf{B}$

If you see → scalar input: use ∇ or ∇^2. Vector input: use $\nabla\cdot$ or $\nabla\times$. Check output type matches.

RADIAL IDENTITIES · MUST MEMORIZE

$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, $r = |\mathbf{r}| = \sqrt{x^2+y^2+z^2}$

$\nabla\cdot\mathbf{r} = 3$ → instant $\quad \nabla\times\mathbf{r} = \mathbf{0}$ → instant $\quad \nabla(r^2) = 2\mathbf{r}$ → instant

$\nabla(r^n) = nr^{n-2}\mathbf{r} = nr^{n-1}\hat{\mathbf{r}}$ $\quad \nabla(1/r) = -\mathbf{r}/r^3 = -\hat{\mathbf{r}}/r^2$

Radial Laplacian: $\nabla^2 f(r) = f''(r) + \dfrac{2}{r}f'(r)$ → critical for $\ln r$, r^n types

If you see $(x^2+y^2+z^2)^n$ or $h(r)$ → convert to radial form $r^k\hat{\mathbf{r}}$ FIRST, then apply radial formula.

2024 auto-marks: $\nabla\cdot\mathbf{r} = 3$, $\nabla\times\mathbf{r} = \mathbf{0}$, $\nabla r^2 = 2\mathbf{r}$. Write these instantly.

Common trap: mixing r and r^2, or writing scalar when vector is needed → **fix:** check output type after every computation.

CURVES · ARCLENGTH · HELIX

$\mathbf{r}(t) = (x(t), y(t), z(t))$, $\mathbf{v} = \mathbf{r}'(t)$, $ds = |\mathbf{r}'(t)|dt$, $s(t) = \int_{t_0}^{t}|\mathbf{r}'(u)|\,du$

Helix: $\mathbf{r}(t) = (a\cos t, a\sin t, bt)$. $\mathbf{r}' = (-a\sin t, a\cos t, b)$. $|\mathbf{r}'| = c = \sqrt{a^2+b^2}$ → constant

If you see $\mathbf{r}(t) = (a\cos t, a\sin t, bt)$ → it is a helix. Execute the FS chain below.

Exam sequence: differentiate $\to$ speed $\to$ arclength $\to$ unit tangent $\to$ unit normal $\to$ binormal $\to$ curvature/torsion.

FRENET-SERRET CORE

$\hat{\mathbf{t}} = \dfrac{d\mathbf{r}}{ds} = \dfrac{\mathbf{r}'}{|\mathbf{r}'|}$ → normalize first

$\kappa = \left|\dfrac{d\hat{\mathbf{t}}}{ds}\right|$

$\dfrac{d\hat{\mathbf{t}}}{ds} = \kappa\hat{\mathbf{n}}$ → then $\hat{\mathbf{n}} = \dfrac{1}{\kappa}\dfrac{d\hat{\mathbf{t}}}{ds}$

$\hat{\mathbf{b}} = \hat{\mathbf{t}}\times\hat{\mathbf{n}}$ → cross product order matters

$\dfrac{d\hat{\mathbf{b}}}{ds} = -\tau\hat{\mathbf{n}}$

Helix results: $\kappa = \dfrac{a}{a^2+b^2}$, $\tau = \dfrac{b}{a^2+b^2}$, $\boldsymbol{\Omega} = \kappa\hat{\mathbf{b}} + \tau\hat{\mathbf{t}}$.

Common trap: computing $d\hat{\mathbf{t}}/dt$ and calling it κ → **fix:** ALWAYS divide by $|\mathbf{r}'|$ to convert $dt \to ds$.

CYLINDRICAL / SPHERICAL OPERATORS

Cyl: $\nabla f = f_r\hat{\mathbf{r}} + \dfrac{1}{r}f_\theta\hat{\boldsymbol{\theta}} + f_z\hat{\mathbf{z}}$

$\nabla\cdot\mathbf{F} = \dfrac{1}{r}\partial_r(rF_r) + \dfrac{1}{r}\partial_\theta F_\theta + \partial_z F_z$ → note $\frac{1}{r}\partial_r(rF_r)$ structure

$\nabla^2 f = \dfrac{1}{r}\partial_r(rf_r) + \dfrac{1}{r^2}f_{\theta\theta} + f_{zz}$

Sph: $\nabla f = f_r\hat{\mathbf{r}} + \dfrac{1}{r}f_\phi\hat{\boldsymbol{\phi}} + \dfrac{1}{r\sin\phi}f_\theta\hat{\boldsymbol{\theta}}$

$\nabla\cdot\mathbf{F} = \dfrac{1}{r^2}\partial_r(r^2F_r) + \dfrac{1}{r\sin\phi}\partial_\phi(\sin\phi\,F_\phi) + \dfrac{1}{r\sin\phi}\partial_\theta F_\theta$

$\nabla^2 f = \dfrac{1}{r^2}\partial_r(r^2 f_r) + \dfrac{1}{r^2\sin\phi}\partial_\phi(\sin\phi\,f_\phi) + \dfrac{1}{r^2\sin^2\phi}f_{\theta\theta}$ → use if f depends on ϕ or θ

Purely radial $\mathbf{F} = F_r(r)\hat{\mathbf{r}}$: $\nabla\cdot\mathbf{F} = \dfrac{1}{r^2}\dfrac{d}{dr}(r^2F_r)$. $\nabla\times\mathbf{F} = \mathbf{0}$ away from singularities.

If you see $(x^2+y^2+z^2)^n(x,y,z)$ → rewrite as $r^{2n+1}\hat{\mathbf{r}}$, identify F_r, use spherical div formula.

Common trap: forgetting $\hat{\mathbf{r}} = \mathbf{r}/r$ introduces extra r → **fix:** write F_r explicitly before applying formula.

COORDINATES · JACOBIANS · BOUNDS

Polar / Cyl: $x = r\cos\theta$, $y = r\sin\theta$, $dA = r\,dr\,d\theta$, $dV = r\,dr\,d\theta\,dz$

Spherical: $x = r\sin\phi\cos\theta$, $y = r\sin\phi\sin\theta$, $z = r\cos\phi$

$dV = r^2\sin\phi\,dr\,d\phi\,d\theta$ → never drop $\sin\phi$

First octant $\Rightarrow 0\le\theta\le\pi/2$, $0\le\phi\le\pi/2$.

Cone $z = r \Rightarrow \phi = \pi/4$.

Sphere radius $a \Rightarrow r = a$.

Offset cylinder $x^2 + (y-1)^2 = 1 \Rightarrow r = 2\sin\theta$.

Setup rule: sketch the shadow region first, then choose coordinates by the geometry trigger table.

TRIPLE-INTEGRAL MICRO-TEMPLATES

Volume: $V(D) = \iiint_D 1\,dV$ → integrand is 1, not the surface

Centroid: $\bar{z} = \dfrac{1}{V}\iiint_D z\,dV$ → divide by V at the end

Cartesian: inner variable runs from lower surface to upper surface. → sketch region first

Cylindrical: sidewall → r bound. Floor → z lower. Roof → z upper. θ from geometry. Attach Jacobian r.

Spherical: sphere → r bound. Cone → ϕ bound. Octant → θ bound. Attach $r^2\sin\phi$.

If you see $\iiint$ → sketch solid, pick coordinates by geometry, write bounds, attach Jacobian.

EP2 Q11 pattern: sidewall $r = 2\sin\theta$, floor $z = 0$, roof $z = r$, $dV = r\,dz\,dr\,d\theta$.

Common trap: writing $dV = dr\,d\theta\,dz$ without Jacobian → **fix:** cyl $\to r$, sph $\to r^2\sin\phi$.

SURFACE PARAMETRIZATION · AREA

Parametric surface: $\mathbf{X}(u,v)$

$dS = |\mathbf{X}_u\times\mathbf{X}_v|\,du\,dv$

Graph surface: $z = f(x,y)$

$\mathbf{X}(x,y) = \langle x, y, f(x,y)\rangle$

$dS = \sqrt{1+f_x^2+f_y^2}\,dx\,dy$ → do not forget the $\sqrt{\ }$

Always find the projection region in the xy-plane before integrating. → sketch region

Paraboloid cap template: solve intersection first. If $z = 2 - r^2$ and cone $z = r$, then $2 - r^2 = r \Rightarrow r = 1$.

Then $A = \int_0^{2\pi}\int_0^1\sqrt{1+4r^2}\,r\,dr\,d\theta$.

If you see "find the area of $z = f(x,y)$" → Monge formula: $\sqrt{1+f_x^2+f_y^2}$, project onto xy-plane.

Common trap: using the bounding surface's derivatives instead of the named surface → **fix:** check which surface the question asks for.

COMMON SURFACE-AREA PATTERNS

Problem type	Minimal skeleton
Graph $z = f(x,y)$	$f_x, f_y \to \sqrt{1+f_x^2+f_y^2}$ → project region → polar if circular
Sphere	parameterize by angles $\to \mathbf{X}_u\times\mathbf{X}_v$
Paraboloid above cone	solve intersection in r first, then Monge formula on paraboloid

LINE INTEGRALS · DIRECT METHOD

$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,dt$ → substitute path into $\mathbf{F}$ first

In 2D: $\int_C P\,dx + Q\,dy$

Workflow: parameterize $C \to$ compute $d\mathbf{r} \to$ write $\mathbf{F}(\mathbf{r}(t))$ explicitly $\to$ dot $\to$ integrate.

LINE-INTEGRAL PATTERNS TO DRILL

Type	Minimal symbolic template
Straight segment	$\mathbf{r}(t) = \mathbf{A} + t(\mathbf{B}-\mathbf{A})$, $0\le t\le 1$
Circle	$\mathbf{r}(t) = (a\cos t, a\sin t)$ or 3D variant
Helix	$\mathbf{r}(t) = (\cos t, \sin t, t)$. $\mathbf{r}' = (-\sin t, \cos t, 1)$
Broken path	split $C = C_1\cup C_2\cup\cdots$, integrate each, sum
Path in 2D	substitute $y = y(x)$, $dy = y'(x)dx$, single-variable integral

2023 MT2 pattern: compute work on a helix, then on a straight segment between the same endpoints and compare. → values will differ

CIRCULATION TEMPLATE · §4.4

Circulation $= \oint_C \mathbf{F}\cdot d\mathbf{r}$: net tangential component of $\mathbf{F}$ around a closed curve. → CCW = positive

Step 1. Parameterize: circle radius $a \to \mathbf{r}(t) = (a\cos t, a\sin t)$, $0\le t\le 2\pi$ → CCW

Step 2. $\mathbf{r}'(t) = (-a\sin t, a\cos t)$

Step 3. Substitute: $\mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)$ → simplify → integrate over $[0, 2\pi]$.

If you see $\oint$ or "circulation" → parameterize closed curve (CCW), compute $\mathbf{r}'$, substitute $\mathbf{F}$, dot, integrate.

Common trap: wrong orientation reverses sign → **fix:** confirm CCW. Reversing direction flips the answer.

TOP 10 EXAM-READY DRILLS

#	Source	Why it matters
1	2025 MT2 Q2(a)–(b)	Best complete radial-field divergence/curl drill
2	2024 MT2 Q1(a)–(c)	Fast operator identities, nearly automatic marks
3	Tutorial 7 Q1(a)–(c)	Derive radial-gradient identities, not just memorize
4	2022 MT2 Q1(b)	Best Frenet-Serret frame drill
5	2025 MT2 Q1	Best graph-surface area exam problem
6	2023 MT2 Q2(a)	Best 3D direct line-integral drill
7	2023 MT2 Q1(c)	Radial divergence with parameter condition
8	Tutorial 7 Q2	Training version of paraboloid-area setup
9	2022 MT2 Q1(c)	Curvature and torsion follow-up after frame mastery
10	Extra Problems 2 Q11(a)	Best cylindrical triple-integral setup drill

COMMON BOUND / GEOMETRY TRIGGERS

Given geometry	Immediate move
x^2+y^2 everywhere	Switch to polar/cylindrical
$x^2+y^2+z^2$ or sphere/cone	Test spherical first
Graph $z = f(x,y)$	Project onto xy-plane, use Monge
Patch inside cylinder	Cylinder defines boundary disk only
Offset circle/cylinder	Rewrite in polar: $r =$ function of θ
First octant	All coords ≥ 0, angle bounds shrink
Helix	Use parametric t, speed is constant

COMMON MISTAKES TO ELIMINATE

Operators: writing scalar where vector is required, or vice versa. → check output type

Radial questions: failing to rewrite field in radial form before differentiating. → convert first

Frenet-Serret: using d/dt as if it were d/ds. → divide by $|\mathbf{r}'|$

Surface area: forgetting projection region or using wrong surface's derivatives. → name the surface

Triple integrals: missing Jacobian r or $r^2\sin\phi$. → write Jacobian immediately

Line integrals: not substituting both path and differential. → write $\mathbf{F}(\mathbf{r}(t))$ first

Circulation: wrong orientation reverses sign. → confirm CCW

Exam habit: before integrating, say silently: formula, bounds, Jacobian, orientation, variable order.

FOLLOW BLINDLY PIPELINES

Line Integral: 1. Parameterize C. 2. Compute $d\mathbf{r}$. 3. Substitute $\mathbf{F}(\mathbf{r}(t))$. 4. Dot and integrate.

Surface Area: 1. Write dS. 2. Choose graph or parametrization. 3. Project region. 4. Integrate.

Triple Integral: 1. Sketch solid. 2. Choose coordinates. 3. Write bounds. 4. Attach Jacobian and integrate.

Goal: automatic operator marks + helix workflow + area setup + direct line-integral workflow.

MH1803 Midterm 2 CHEATSHEET MODE: Worked Micro-Templates

Use this side for method execution.

Keep the algebra linear: write formula first, substitute second, simplify last.

OPERATOR TEMPLATE A · INSTANT IDENTITIES

1. $\nabla\cdot\mathbf{r}=1+1+1=3$ **2.** $\nabla\times\mathbf{r}=\mathbf{0}$ (all cross-partials cancel) **3.** $\nabla r^2=2\mathbf{r}$

Derivation of 3: $\nabla(x^2+y^2+z^2)=(2x,2y,2z)=2\mathbf{r}$. → write these three instantly for free marks

Use on: 2024 MT2 Q1(a)–(c), Tutorial 7 Q1(a), Lecture 3A operator warm-ups.

OPERATOR TEMPLATE B · RADIAL FIELD

If $\mathbf{F}=(x^2+y^2+z^2)^n(x,y,z)$, then

$\mathbf{F}=r^{2n}\mathbf{r}=r^{2n+1}\hat{\mathbf{r}}$ → identify $F_r=r^{2n+1}$

Divergence route:

$$\nabla\cdot\mathbf{F}=\frac{1}{r^2}\frac{d}{dr}(r^2F_r)=\frac{1}{r^2}\frac{d}{dr}(r^{2n+3})=(2n+3)r^{2n}$$

Solenoidal condition: $(2n+3)r^{2n}=0\Rightarrow n=-\frac{3}{2}$ (away from singularities).

Curl route: pure radial field $\Rightarrow\nabla\times\mathbf{F}=\mathbf{0}$ away from singularities. → state this directly

Common trap: losing the extra r when converting $\mathbf{r}$ to $\hat{\mathbf{r}}$ → **fix:** $\mathbf{r}=r\hat{\mathbf{r}}$, so $r^{2n}\mathbf{r}=r^{2n+1}\hat{\mathbf{r}}$.

OPERATOR TEMPLATE C · $\nabla(r^n)$ AND $\nabla(1/r)$

$r^n=(x^2+y^2+z^2)^{n/2}$

$\partial_x(r^n)=\frac{n}{2}(x^2+y^2+z^2)^{n/2-1}(2x)=nxr^{n-2}$

Similarly, $\partial_y(r^n)=nyr^{n-2}$, $\partial_z(r^n)=nzr^{n-2}$ → same pattern for all

So $\nabla(r^n)=nr^{n-2}(x,y,z)=nr^{n-2}\mathbf{r}$

Set $n=-1$: $\nabla(1/r)=-r^{-3}\mathbf{r}=-\hat{\mathbf{r}}/r^2$

Use on: Tutorial 7 Q1(b)–(c), Lecture 3A Example 3.17, radial simplifications in past papers.

OPERATOR TEMPLATE D · $\nabla\times(\mathbf{k}\times\mathbf{r})$ [2023 Q1(a)]

Identity: $\nabla\times(\mathbf{A}\times\mathbf{B})=\mathbf{A}(\nabla\cdot\mathbf{B})-\mathbf{B}(\nabla\cdot\mathbf{A})+(\mathbf{B}\cdot\nabla)\mathbf{A}-(\mathbf{A}\cdot\nabla)\mathbf{B}$

Step 1. $\mathbf{k}$ constant $\Rightarrow\nabla\cdot\mathbf{k}=0$, $(\mathbf{r}\cdot\nabla)\mathbf{k}=\mathbf{0}$

Step 2. $(\mathbf{k}\cdot\nabla)\mathbf{r}=\mathbf{k}$ → since $\partial_i r_j=\delta_{ij}$

Step 3. $\nabla\times(\mathbf{k}\times\mathbf{r})=\mathbf{k}(\nabla\cdot\mathbf{r})-\mathbf{0}+\mathbf{0}-\mathbf{k}=3\mathbf{k}-\mathbf{k}=2\mathbf{k}$

General rule: $\nabla\times(\mathbf{c}\times\mathbf{r})=2\mathbf{c}$ **for ANY constant vector** $\mathbf{c}$**. If you see** $\nabla\times(\mathbf{k}\times\mathbf{r})$ **→ answer is** $2\mathbf{k}$ **instantly.**

OPERATOR TEMPLATE E · LAPLACIAN OF $\ln(r^2)$ [2023 Q1(b)]

$\ln(x^2+y^2+z^2)=\ln(r^2)=2\ln r$. Set $f(r)=2\ln r$.

Step 1. $f'(r)=2/r$, $f''(r)=-2/r^2$

Step 2. $\nabla^2 f=f''(r)+\frac{2}{r}f'(r)=-\frac{2}{r^2}+\frac{4}{r^2}=\frac{2}{r^2}$

If you see $\nabla^2(\ln(x^2+y^2+z^2))$ **→ rewrite as** $2\ln r$**, apply radial Laplacian, answer is** $2/r^2$**.**

OPERATOR TEMPLATE F · PRODUCT-RULE APPLICATION

Rules: $\nabla\cdot(u\mathbf{F})=u\nabla\cdot\mathbf{F}+\mathbf{F}\cdot\nabla u$. $\nabla\times(u\mathbf{F})=\nabla u\times\mathbf{F}+u\nabla\times\mathbf{F}$

Example: $\nabla\cdot(r^2\mathbf{r})$. Set $u=r^2$, $\mathbf{F}=\mathbf{r}$.

Step 1. $\nabla u=\nabla(r^2)=2\mathbf{r}$. $\nabla\cdot\mathbf{r}=3$ → instant facts

Step 2. $\nabla\cdot(r^2\mathbf{r})=r^2(3)+\mathbf{r}\cdot(2\mathbf{r})=3r^2+2r^2=5r^2$

If you see $\nabla\cdot(f\mathbf{F})$ **or** $\nabla\times(f\mathbf{F})$ **→ apply product rule, then use instant identities for each piece.**

HELIX TEMPLATE A · COMPLETE FS CHAIN + DARBOUX

Step 1. $\mathbf{r}(t)=(a\cos t,a\sin t,bt)$. $\mathbf{r}'(t)=(-a\sin t,a\cos t,b)$

Step 2. $|\mathbf{r}'|=c=\sqrt{a^2+b^2}$ → constant

Step 3. $\hat{\mathbf{t}}=\frac{1}{c}(-a\sin t,a\cos t,b)$ → normalize

Step 4. $\frac{d\hat{\mathbf{t}}}{dt}=\frac{1}{c}(-a\cos t,-a\sin t,0)$

Step 5. $\frac{d\hat{\mathbf{t}}}{ds}=\frac{1}{c^2}(-a\cos t,-a\sin t,0)$ → divide by c

Step 6. $\kappa=\frac{a}{c^2}$. $\hat{\mathbf{n}}=(-\cos t,-\sin t,0)$

Step 7. $\hat{\mathbf{b}}=\hat{\mathbf{t}}\times\hat{\mathbf{n}}=\frac{1}{c}(b\sin t,-b\cos t,a)$

Step 8. Torsion: $\frac{d\hat{\mathbf{b}}}{ds}=\frac{1}{c^2}(b\cos t,b\sin t,0)=-\tau\hat{\mathbf{n}}$ → $\tau=\frac{b}{c^2}$. **Darboux:** $\boldsymbol{\Omega}=\kappa\hat{\mathbf{b}}+\tau\hat{\mathbf{t}}$

Memorize: $\kappa=a/c^2$, $\tau=b/c^2$ where $c=\sqrt{a^2+b^2}$. **Darboux:** substitute $\kappa,\tau,\hat{\mathbf{b}},\hat{\mathbf{t}}$ from above.

Numerical check (a=3, b=4): $c=5$. $\hat{\mathbf{t}}=\frac{1}{5}(-3\sin t,3\cos t,4)$. $\kappa=3/25$. $\tau=4/25$.

SURFACE-AREA TEMPLATE A · GRAPH SURFACE

If $z=f(x,y)$, then $A=\iint_R\sqrt{1+f_x^2+f_y^2}\,dA$ → Monge formula

Paraboloid: $f(x,y)=x^2+y^2$

Step 1. $f_x=2x$, $f_y=2y$

Step 2. $\sqrt{1+4x^2+4y^2}=\sqrt{1+4r^2}$ → convert to polar

Step 3. If $0\le z\le 2$, then $x^2+y^2\le 2$, so $r\le\sqrt{2}$ → projection region

Step 4. $A=\int_0^{2\pi}\int_0^{\sqrt{2}}\sqrt{1+4r^2}\,r\,dr\,d\theta$

Use on: Tutorial 7 Q2 and as training for 2025 MT2 Q1.

SURFACE-AREA TEMPLATE B · 2025 CAP

Step 1. Paraboloid: $z=2-r^2$. Cone: $z=r$.

Step 2. Intersection: $2-r^2=r\Rightarrow r^2+r-2=0\Rightarrow r=1$ → find intersection first

Step 3. $f_x=-2x$, $f_y=-2y$ → derivatives of the paraboloid

Step 4. $\sqrt{1+4x^2+4y^2}=\sqrt{1+4r^2}$

Step 5. $A=\int_0^{2\pi}\int_0^1\sqrt{1+4r^2}\,r\,dr\,d\theta$

Common trap: using cone derivatives instead of paraboloid derivatives → **fix:** the area is of the paraboloid cap, not the cone.

LINE-INTEGRAL TEMPLATE A · HELIX

Step 1. $\mathbf{r}(t)=(\cos t,\sin t,t)$. $\mathbf{r}'(t)=(-\sin t,\cos t,1)$

Step 2. For $\mathbf{F}=(-y,x,z)$: $\mathbf{F}(\mathbf{r}(t))=(-\sin t,\cos t,t)$ → substitute path into $\mathbf{F}$

Step 3. $\mathbf{F}\cdot\mathbf{r}'=\sin^2 t+\cos^2 t+t=1+t$ → simplifies!

Step 4. $W=\int_a^b(1+t)\,dt$

Lesson: after substitution, ugly 3D integrals often collapse to simple scalar integrands.

LINE-INTEGRAL TEMPLATE B · STRAIGHT SEGMENT

Step 1. From A to B: $\mathbf{r}(t)=A+t(B-A)$, $0\le t\le 1$

Step 2. $d\mathbf{r}=(B-A)\,dt$ → constant direction

Step 3. $\int_C\mathbf{F}\cdot d\mathbf{r}=\int_0^1\mathbf{F}(\mathbf{r}(t))\cdot(B-A)\,dt$

Common trap: assuming same endpoints $\Rightarrow$ same value → **fix:** without conservative-field theorem (out of scope), different paths give different values.

LINE-INTEGRAL TEMPLATE C · 2D DIFFERENTIAL FORM

$\int_C P\,dx+Q\,dy$: substitute $y=y(x)$, $dy=y'(x)dx$ → $\int_a^b(P+Qy')dx$ → single variable

Broken path: integrate each segment separately and add. **Circle:** $x=a\cos t$, $y=a\sin t$. → check orientation

CIRCULATION TEMPLATE · WORKED EXAMPLE

Example: $\mathbf{F}=(-y,x)$, C: unit circle CCW.

Step 1. $\mathbf{r}(t)=(\cos t,\sin t)$, $0\le t\le 2\pi$. $\mathbf{r}'=(-\sin t,\cos t)$

Step 2. $\mathbf{F}(\mathbf{r}(t))=(-\sin t,\cos t)$

Step 3. $\mathbf{F}\cdot\mathbf{r}'=\sin^2 t+\cos^2 t=1$

Step 4. $\oint=\int_0^{2\pi}1\,dt=2\pi$

Lesson: after substitution, circulation integrals often simplify to constants.

TRIPLE-INTEGRAL TEMPLATE A · NESTED CARTESIAN

$\iiint f(x,y,z)\,dz\,dx\,dy$

Rule: integrate inside first, substitute its bounds completely, then move outward. → inside out

Never: carry the wrong variable into the next stage after evaluating bounds.

TRIPLE-INTEGRAL TEMPLATE B · CYLINDRICAL REGION

Step 1. Region: inside $x^2+(y-1)^2=1$, above $z=0$, below $z=\sqrt{x^2+y^2}$

Step 2. Sidewall: $x^2+y^2-2y=0\Rightarrow r=2\sin\theta$ → sidewall gives r bound

Step 3. Floor $z=0$, roof $z=r$ → floor/roof give z bounds

Step 4. $0\le\theta\le\pi$, $0\le r\le 2\sin\theta$, $0\le z\le r$

Step 5. $V=\int_0^\pi\int_0^{2\sin\theta}\int_0^r 1\cdot r\,dz\,dr\,d\theta$ → the r is the Jacobian

$\bar{z}=\frac{1}{V}\int_0^\pi\int_0^{2\sin\theta}\int_0^r z\,r\,dz\,dr\,d\theta$ → divide by V at end

Common trap: the extra r in the integrand is the Jacobian, not from the roof $z=r$ → **fix:** write Jacobian separately.

TRIPLE-INTEGRAL TEMPLATE C · SPHERICAL BOUNDS

Sphere: $r=a$

Cone: $z=r\Rightarrow\phi=\pi/4$

First octant: $0\le\theta\le\pi/2$, $0\le\phi\le\pi/2$

$dV=r^2\sin\phi\,dr\,d\phi\,d\theta$ → never drop $r^2\sin\phi$

Use on: Extra Problems 2 Q8 and any sphere/cone bound-construction question.

LAST-48-HOUR PRACTICE ORDER

Block	Problems	Purpose
1	2024 MT2 Q1(a)–(c), 2023 MT2 Q1(a)–(b), Tut 7 Q1	Operators + cross-product + Laplacian
2	2025 MT2 Q2(a)–(b), 2023 MT2 Q1(c)	Radial-field recognition
3	2022 MT2 Q1(a)–(c)	Helix + FS chain
4	Tut 7 Q2, 2025 MT2 Q1	Graph-surface area
5	2023 MT2 Q2(a)–(b), Tut 8 Q1	Line integrals / path comparison
6	EP2 Q8, Q11(a)	3D bounds and Jacobians

30-SECOND PRE-SUBMISSION CHECK

1. Did I choose the correct formula/operator first?
2. Are my bounds matched to the picture and coordinate system?
3. Did I include the Jacobian r or $r^2\sin\phi$?
4. For FS, did I convert d/dt to d/ds?
5. For line integrals, did I substitute both the path and $d\mathbf{r}$?
6. For circulation, is my orientation correct (CCW = positive)?
7. Did I accidentally use any out-of-scope theorem shortcut?